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Understanding a Netbook Desktop

Desktop environments are wonderful in the way they just work. The complexity of what is happening 'below the hood' is well hidden from us. Looking at the various netbook options, I was curious to understand what it involved to create or modify a netbook desktop experience.

The idea is simple. Once you sign in, a start-up program for the session runs the desired window manager, and typically displays the desktop icons using a file manager, and starts a panel. It then runs the components that need to be auto-started.

However, trying to understand the detailed flow reminded me of a comment I read many years ago, when I was trying to understand an Algol source file. The comment at the entry point said: "This is where you start. Where you end up is your problem!"

Let us then start at where we end up. You are more likely to be interested in controlling the components or applications that are run as soon as the session is started. A standard location for these is */etc/xdg/autostart*. A desktop environment may, of course, change it. Anyone interested in more details can see the standards proposed by *freedesktop.org* (http://www.freedesktop.org/wiki/Specifications). Since you are likely to use a GUI, you don't really need the name of this directory.

The keywords '*OnlyShowIn*' and '*NotShowIn*' can control the desktop environments for which a component is applicable. You can easily control the start-up applications for your environment by using a preferences option like *Startup Applications* in Ubuntu Netbook Edition, or on the standard GNOME desktop. The user-modified files will be in the *.config/autostart* directory in your home directory.

I was, however, interested in understanding what happens before you reach this point. A convenient starting point is the display manager. We typically use GDM or KDM. The display manager will usually rely on desktop files in */usr/share/xsessions* and */etc/X11/sessions* to offer a list of options available. The former is the common usage now, in major distributions including Fedora, Ubuntu and SuSE.

MeeGo

MeeGo used its own display manager cum session launcher, *uxlaunch*. The USP of this launcher is that it starts the X server as a user process, not owned by the root. The disadvantage is that it is a single-user system, at least at present. A desktop session will be started for the default user. There is no option to log out or log in as a different user.

This is not a strict requirement for running MeeGo. Smeegol by SuSE, based on MeeGo, uses GDM—and a MeeGo desktop is one of the options possible. GDM is the display manager, which runs *uxlaunch* to start a MeeGo session.

Ubuntu Unity

On Ubuntu Unity, GDM has 'Ubuntu Desktop Edition' and 'Ubuntu Netbook Edition' (UNE) as separate options. However, the desktop files for both show that the same *gnome-session* is run. The essential difference between the two is the name of the desktop session.

For a standard GNOME session, *gconf/schemas/gnome-session.schemas* defines the various options, including the required components: window manager, file manager and panel. In the case of Ubuntu, *gnome-session* uses the files in */var/lib/gconf*. These are actually generated from the files in */usr/share/gconf*. The package '*ubuntu-netbook-default-settings*' installs *une/mandatory* and *une/defaults*, which override the default values in case the desktop session is UNE. The window manager is replaced by '*mutter*', the panel by a null string (which is indeed reported as an error in *.xsession-errors*). The option to display the desktop icons by Nautilus is set to false. A critical specification is the clutter plug-in to be used by '*mutter*'. For the Ubuntu Unity interface, it is '*libunity-mutter*'. This plug-in controls the panels on the top and at the side, and the launching of various applications.

Mutter is also the window manager used by MeeGo and Smeegol. It uses '*moblin-netbook*' as the *clutter* plug-in. It replaces the conventional panels with a tabbed view of the same functionality. Mutter is a fork of Metacity, with Clutter support. It is likely to be the default window manager for GNOME 3, as GNOME Shell is integrated with Mutter. (http://en.wikipedia.org/wiki/GNOME_Shell).

KDE Netbook

In the case of KDE Netbook Workspace, GDM or KDM will run 'startkde', which will use your workspace preference settings to show the appropriate desktop. KDE follows a more integrated approach. The user experience is controlled by containers (containments), which contain plug-ins, applets, applications, etc. Common types of containments are the panel and the desktop. A starting environment for a distribution is packaged and provided by one or more packages containing the settings—for example, *kubuntu-settings* and *kubuntu-netbook-*